

He Min

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EDUCATIONAL BACKGROUND

Nanyang Technological University

Master of Science in Signal Processing and Machine Learning

Chongqing University of Technology

Bachelor of Science in Computer Science and Technology (Honours)

• GPA: 3.96 / 5.0 (Top 1%)

Singapore

2026.01-2027.07

Chongqing, China

2021.09-2025.06

INTERNSHIP EXPERIENCE

Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm)

2026.05-2026.08

Research Intern in Embodied AI for Robotics [[github](#)]

Project Overview: Using the Lenovo Morningstar MX robot as a platform, an intelligent agent system, AgenticVLN, was constructed, which is a collaborative system of NavAgent (the task planning mastermind) and DualVLN (a slow-fast dual-system navigation). NavAgent, based on a LangGraph state machine, is responsible for global task parsing, memory storage, telemetry self-checking, and anomaly diagnosis. DualVLN is responsible for specific motion generation, using System2 (2Hz, fine-tuned QwenVL-2.5) to predict 2D pixel sub-targets, and System1 (30Hz, NextDiT) to output high-frequency physical trajectories. SR increased by 10.1%, SPL increased by 6.7%, NE decreased by 17.6%, and StR decreased by 32.5%.

PROJECT EXPERIENCE

DirectVLA: Development of a High-Frequency, Low-Memory Real-Time VLA Strategy for a Simplified Direct Mapping Architecture [[github](#)]

2026.06- Present

Project Overview: Addressing the bottlenecks of traditional VLA, which relies on LLM with billions of parameters, resulting in large memory requirements (>10 GB) and high latency (~11 Hz), we spearheaded a real-time strategy with a minimalist direct mapping architecture. The project abandons the $V \rightarrow L \rightarrow A$ path, constructing a $V+L \rightarrow A$ model based on DINOv3 and BERT, combining bidirectional text-to-image interaction with nonautoregressive continuous action block decoding. On a single RTX 4090, the model achieves an ultra-low latency of ~30ms with only 0.2B parameters and 0.9GB of memory. High robustness closed-loop verification was completed on a single LIBERO arm (97.7%), RoboTwin dual arms (60.2%), and the xMate SR3 real-world device (up to 92.5%).

OmniVLN: Omnidirectional 3D Perception and Token-Efficient LLM Reasoning for Visual-Language Navigation across Air and Ground Platforms

Co-first Author, **IEEE RA-L (SCI Q2 TOP, Under Review)** [[arXiv](#)] Supervisor: Prof. Lihua Xie 2025.12-2026.03

Project Overview: To address the challenges of spatial reasoning deficiencies and context bloat in embodied agents, an autonomous navigation agent architecture based on a LangGraph state graph and a 5-layer dynamic scene graph environment world model is constructed. The system integrates continuous coherent topology-based room allocation and VLM dual-graph pruning to refine spatial representations. At the decision-making end, an innovative ego-centric 3D 8-neighborhood observation model and a multi-resolution spatial attention cue mechanism are implemented, combined with Actor-Critic self-reflection and spatial toolchain enhancements to achieve a long-range zero-shot proactive perception closed loop. Experiments show that the agent's decision success rate reaches 93.18% (with a 22.73% improvement in viewpoint-dependent accuracy) and inference token consumption is reduced by 69.98%.

PUBLICATION

He, M. (2025). A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis. In Proceedings of E-commerce and Artificial Intelligence (ECAI 2024). [[SciTePress](#)]

AWARDS & HONORS

National Scholarship (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

TECHNOLOGY STACK

- **VLA and Embodied Policies:** VLA, ACT, Diffusion Policy, Stream Matching, Sim2Real, LeRobot
- **Spatial Awareness and Embodied Navigation:** VLN, 3D Dynamic Scene Graph, Topological Semantic Map, SAM2, Open3D, KDTree
- **Large Model Agent Architectures:** LangGraph, MCP, Multi-Agent, ReAct, Function Calling, CoT, RAG, Prompt, Skill
- **Simulation Operations and Systems Engineering:** ROS 2, Isaac Sim/MuJoCo/Genesis, C++, Python, Linux, Docker, Git